

LOGISTICS & FLEET ANALYSIS DASHBOARD

Data Analyst Portfolio Project

Tools: Microsoft Excel | Power Query | Power BI | DAX

1. Project Overview

This project analyzes logistics and fleet operations data to evaluate operational efficiency, driver performance, delivery performance, operating costs, and profitability.

The project uses Microsoft Excel, Power Query, Power BI, and DAX to transform raw logistics data into meaningful dashboards and actionable business insights.

2. Business Objective

The objective of this project is to help a logistics company:

- Monitor fleet utilization and operational performance
 - Track fuel and maintenance expenses
 - Evaluate driver performance
 - Monitor delivery efficiency
 - Understand operating costs
 - Identify factors affecting profitability
 - Support data-driven business decisions
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3. Dataset

The project combines multiple logistics datasets covering:

- Drivers
 - Trucks
 - Trailers
 - Customers
 - Facilities
 - Routes
 - Loads
 - Trips
 - Fuel Purchases
 - Maintenance Records
 - Delivery Events
 - Safety Incidents
 - Driver Monthly Metrics
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4. Data Cleaning & Transformation

The data was prepared and transformed using Power Query.

Key data preparation activities included:

- Removing duplicate records
 - Handling missing values
 - Cleaning inconsistent data
 - Standardizing data formats
 - Formatting dates
 - Reviewing and correcting data types
 - Transforming columns
 - Preparing datasets for analysis
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5. Data Modeling

A relational data model was created in Power BI to connect the different logistics datasets.

Key relationships included:

- Loads → Customers
- Loads → Routes
- Trips → Loads
- Trips → Drivers
- Trips → Trucks
- Trips → Trailers
- Fuel Purchases → Trips
- Maintenance Records → Trucks
- Delivery Events → Trips
- Safety Incidents → Trips

This model allowed information from different operational areas to be analyzed together.

6. Tools & Technologies

Microsoft Excel

Used for data storage, initial data preparation, and reviewing datasets.

Power Query

Used for data cleaning, transformation, and preparation.

Power BI

Used for data modeling, visualization, KPI development, and interactive dashboard creation.

DAX

Used to create analytical measures and calculate key performance indicators.

7. Dashboard 1 — Fleet Operations

The Fleet Operations dashboard provides an overview of fleet and operational performance.

Key Areas Analyzed

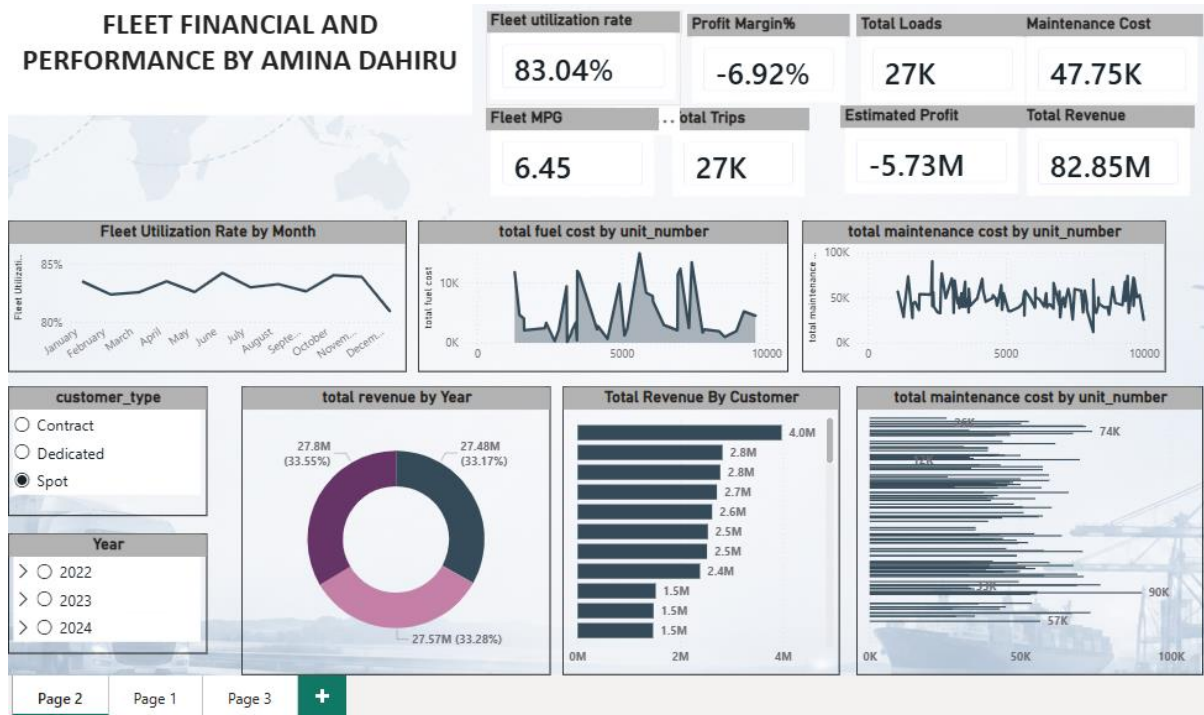
- Fleet utilization
- Fuel costs
- Truck performance
- Operating activity
- Maintenance costs
- Operational KPIs

Key KPIs

- Fleet Utilization Rate
- Total Fuel Cost
- Maintenance Cost

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FLEET FINANCIAL AND PERFORMANCE BY AMINA DAHIRU



8. Dashboard 2 — Driver & Delivery Performance

The Driver & Delivery Performance dashboard evaluates driver activity and delivery performance.

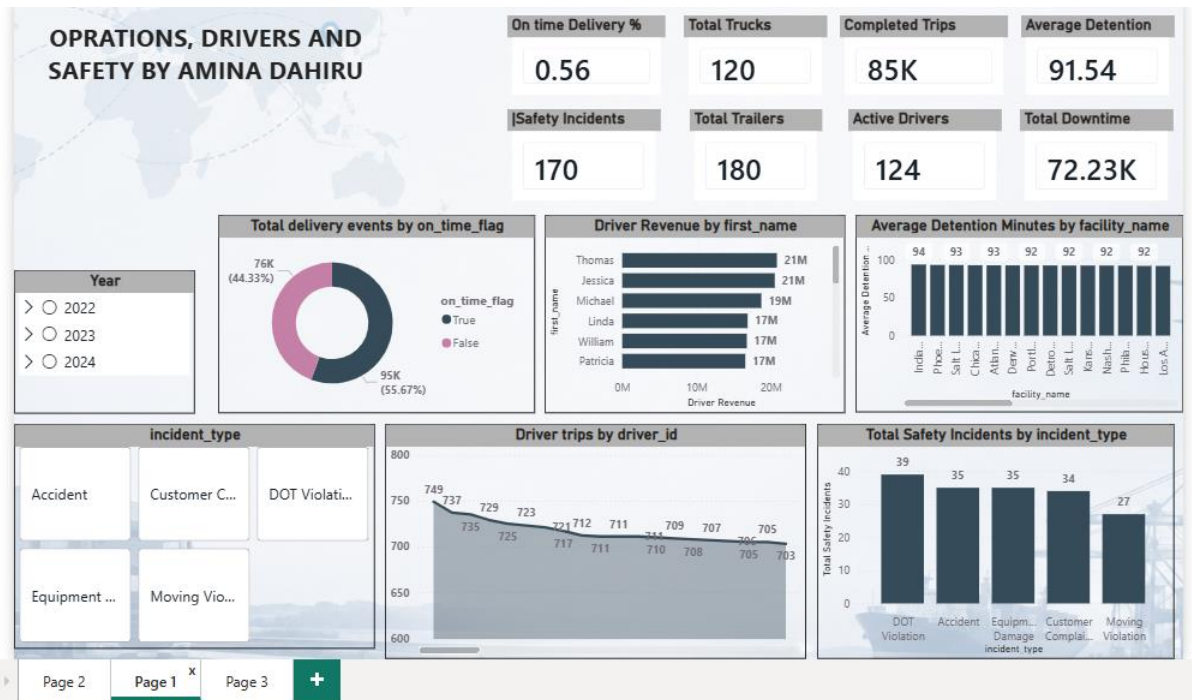
Key Areas Analyzed

- Driver trips
- Driver performance
- On-time delivery
- Delivery events
- Operational efficiency

Key KPIs

- Driver Trips
- On-Time Delivery Rate
- Driver Performance Metrics

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9. Dashboard 3 — Cost & Profitability

The Cost & Profitability dashboard focuses on the financial performance of logistics operations.

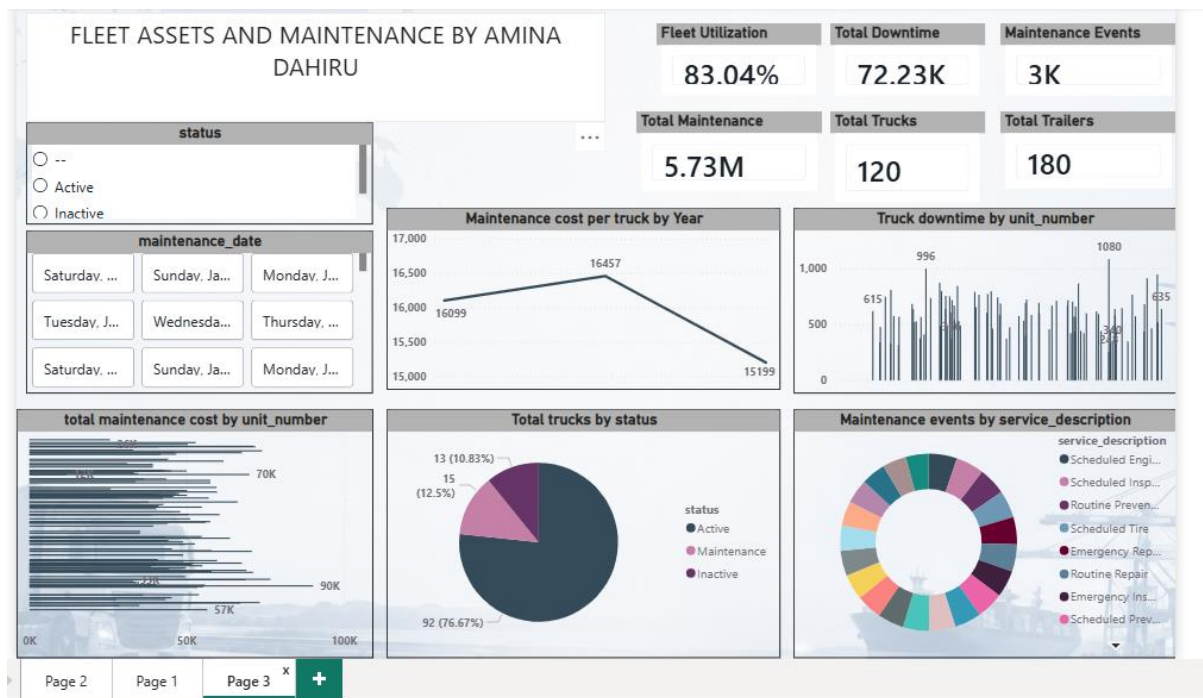
Key Areas Analyzed

- Fuel expenses
- Maintenance costs
- Operating costs
- Revenue
- Estimated profit

Key KPIs

- Total Operating Cost
- Total Fuel Cost
- Maintenance Cost
- Estimated Profit

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10. Key Skills Demonstrated

This project demonstrates practical experience in:

- Data cleaning
- Data transformation
- Data modeling
- Power Query
- DAX
- KPI development
- Dashboard development
- Data visualization
- Business analysis
- Data storytelling
- Insight generation

11. Business Insights

The dashboards were designed to help management identify:

- Areas of high fleet utilization
- Major operational cost drivers
- Fuel expenditure patterns
- Differences in driver performance
- Delivery performance trends
- Maintenance-related expenses
- Factors affecting profitability

12. Business Recommendations

Based on the analysis, logistics management can:

- Monitor fleet utilization regularly
- Identify opportunities to reduce fuel costs
- Track vehicle maintenance expenses
- Improve driver performance monitoring
- Investigate delivery delays

- Monitor operating costs against revenue
 - Focus on areas negatively affecting profitability
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13. Project Outcome

The project demonstrates how raw logistics data can be transformed into an interactive business intelligence solution.

The resulting dashboards provide management with a centralized view of fleet operations, driver and delivery performance, costs, and profitability.

This enables more informed decision-making and helps identify opportunities to improve operational efficiency and financial performance.

14. About the Analyst

I am an aspiring Data Analyst with a background in Cybersecurity and practical experience in data analysis, data cleaning, Microsoft Excel, Power Query, Power BI, DAX, data modeling, and dashboard development.

I enjoy transforming raw data into meaningful insights that can support business decisions and improve operational performance.

15. Tools Used

Microsoft Excel | Power Query | Power BI | DAX

Project Type

End-to-End Data Analysis & Business Intelligence Project